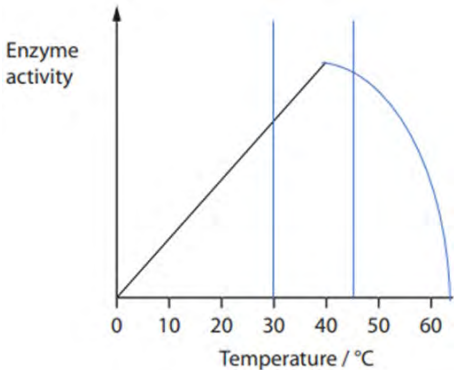


Question number	Answer	Additional guidance	Mark
8(a)(i)	An answer that includes the following points: - linear line that increases with temperature and then decreases (1) - optimum temperature shown at (about) 40 °C (1)	ACCEPT carefully hand-drawn line sloping up to optimum NB line does not have to start at the origin NB optimum must be above 30°C and below 45°C 	(2)

Question number	Answer	Additional guidance	Mark
8(a)(ii)	- values at 30 °C and 40 °C chosen (1) - rate calculated per minute for each (1) - numbers substituted into equation (1) - correct answer to 2 / 3 sig figs with no units (1) OR - values at 40 °C and 50 °C chosen - rate calculated per minute for each (1) - numbers substituted into equation (1) - correct answer to 2 / 3 sig figs with no units (1)	50 and 32.5 and 50 and 22.5 $(50-32.5) \div 5 / 17.5 \div 5 / 3.5$ and $(50-22.5) \div 5 / 27.5 \div 5 / 5.5$ CE from reasonable values read from graph $5.5 \div 3.5 / 1.5714$ CE if added 10 to a correct rate from mp 2 1.6 / 1.57 ACCEPT correctly rounded value for mp 3 OR 50 and 22.5 and 50 and 47.5 $(50-22.5) \div 5 / 27.5 \div 5 / 5.5$ and $(50-47.5) \div 2.5 / 2.5 \div 5 / 0.5$ CE from reasonable values read from graph $0.5 \div 5.5 / 0.090909$ CE if added 10 to a correct rate from mp 2 0.091 / 0.0909 ACCEPT correctly rounded value for mp 3	(4)

Question number	Answer	Additional guidance	Mark										
8(b)(i)	- mass of monosaccharides calculated to be 1361.3 / correct ratios expressed with more than one dp (1) - ratios shown correctly as either (all) whole numbers or both disaccharide and tetrasaccharide values correctly rounded to one dp (1)	NB check working out in space and first table for values eg 164.012, 6.868, 1, 1.145 disaccharide 6.9 and tetrasaccharide 1.1 <table><tr><th>Type of carbohydrate</th><th>Ratio</th></tr><tr><td>monosaccharide</td><td>164</td></tr><tr><td>disaccharide</td><td>7</td></tr><tr><td>trisaccharide</td><td>1</td></tr><tr><td>tetrasaccharide</td><td>1</td></tr></table>	Type of carbohydrate	Ratio	monosaccharide	164	disaccharide	7	trisaccharide	1	tetrasaccharide	1	(2)
Type of carbohydrate	Ratio												
monosaccharide	164												
disaccharide	7												
trisaccharide	1												
tetrasaccharide	1												

Question number	Answer	Mark
8(b)(ii)	The only correct answer is D. <i>A is incorrect because inositol and raffinose have the modal concentration</i> <i>B is incorrect because inositol and raffinose have the modal concentration</i> <i>C is incorrect because inositol and raffinose have the modal concentration</i>	(1)

Question number	Answer	Additional guidance	Mark
8(b)(iii)	General description • bacteria and fungi are decomposers • decomposers release enzymes for decomposition • digested molecules are absorbed into the decomposer • release carbon dioxide Details of decomposition of carbohydrates • carbohydrases are needed to breakdown carbohydrates • hydrolysis of glycosidic bonds • to form monosaccharides • monosaccharides taken up by diffusion • glucose used in respiration • carbon dioxide produced by respiration of glucose Specific detail relating to carbohydrates in the table • one glycosidic bond broken in disaccharides, two in trisaccharides, three in tetrasaccharides	IGNORE references to starch and amylase	

- using one / two / three water molecules
- e.g. sucrose broken down into glucose and fructose
- monosaccharides do not need breaking down

Level 1 : general description of decomposition

1 mark : one point made

2 marks : three points made

Level 2 : details about decomposition of carbohydrates

3 marks : two points made

4 marks : three points made

**Level 3 : specific details relating to carbohydrates in the table
+ mention of respiration releasing carbon dioxide**

5 marks : specific detail given for either di / tri / tetra saccharides

6 marks : specific detail given for two of the three types of
saccharides

(6)

